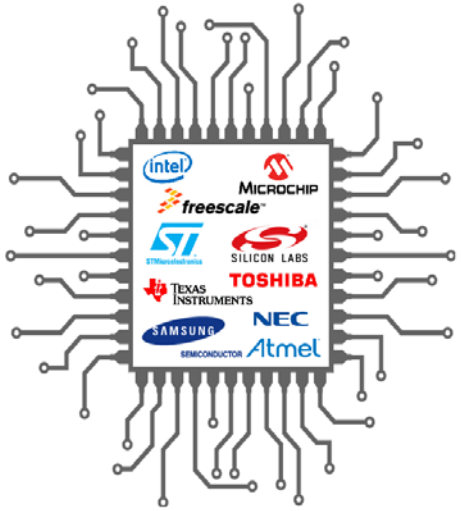
Pulse Width Modulation



Thareendhra Keerthi Wijayasiriwardhane

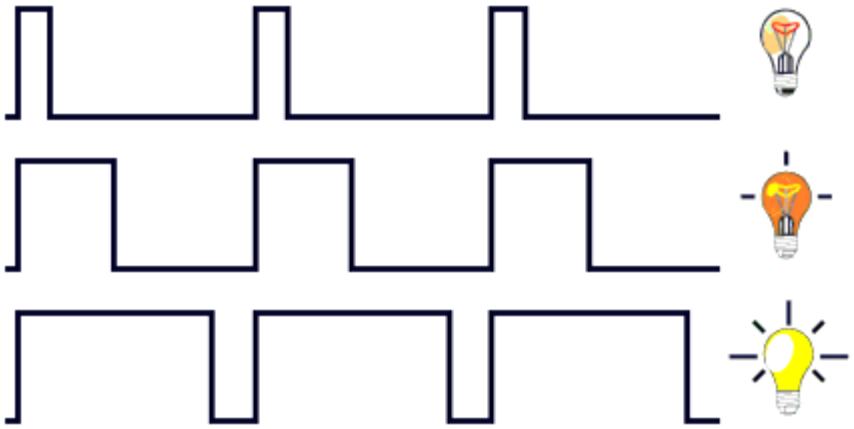
Department of Industrial Management
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Learning Outcomes

- On completion of this session, you should be able to
  - Explain the need of PWM
  - Compute duty cycle and frequency for PWM outputs
  - Describe how Timers generate PWM outputs
  - Explain the functionality of Timer/Counter registers used for PWM
  - Construct a microcontroller-based system and program it to generate a PWM output with a varying duty cycle

Pulse Width Modulation

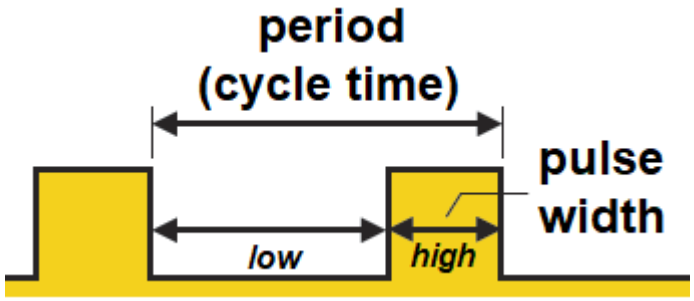
- A technique for getting an analog like output by means of digital
- Toggles a digital output fast enough, making any attached device to react not to the individual pulses, but to the % of time that the output is high in one cycle



  - How fast?

Duty Cycle

- Percentage of time that output is high in one cycle

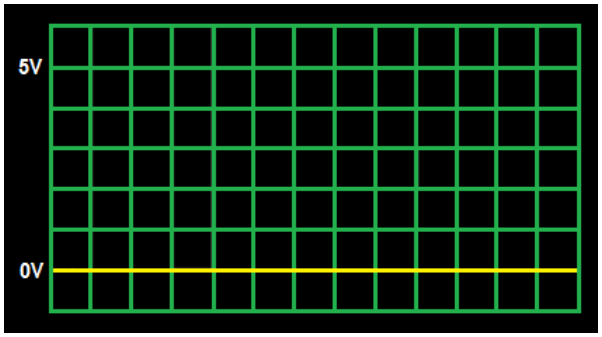


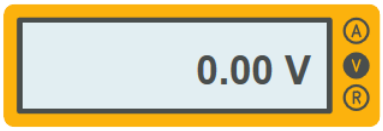
$$duty\ cycle = \frac{t_{high}}{t_{high} + t_{low}} \times 100\%$$

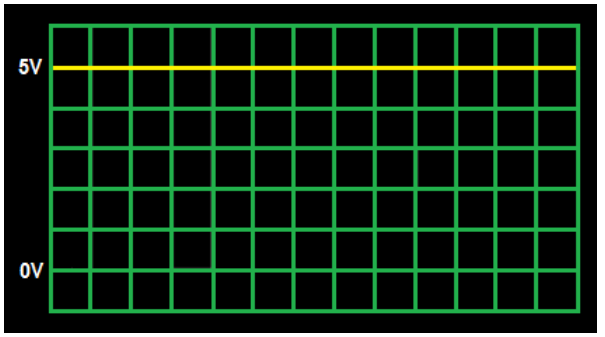
$$= \frac{t_{high}}{period} \times 100\%$$

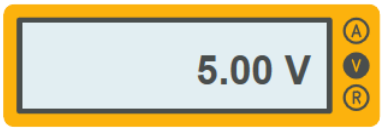
$$= \frac{pulse\ width}{period} \times 100\%$$

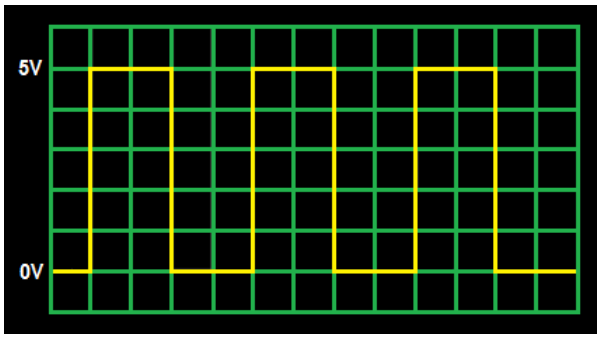
Average Voltage

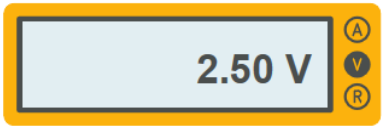






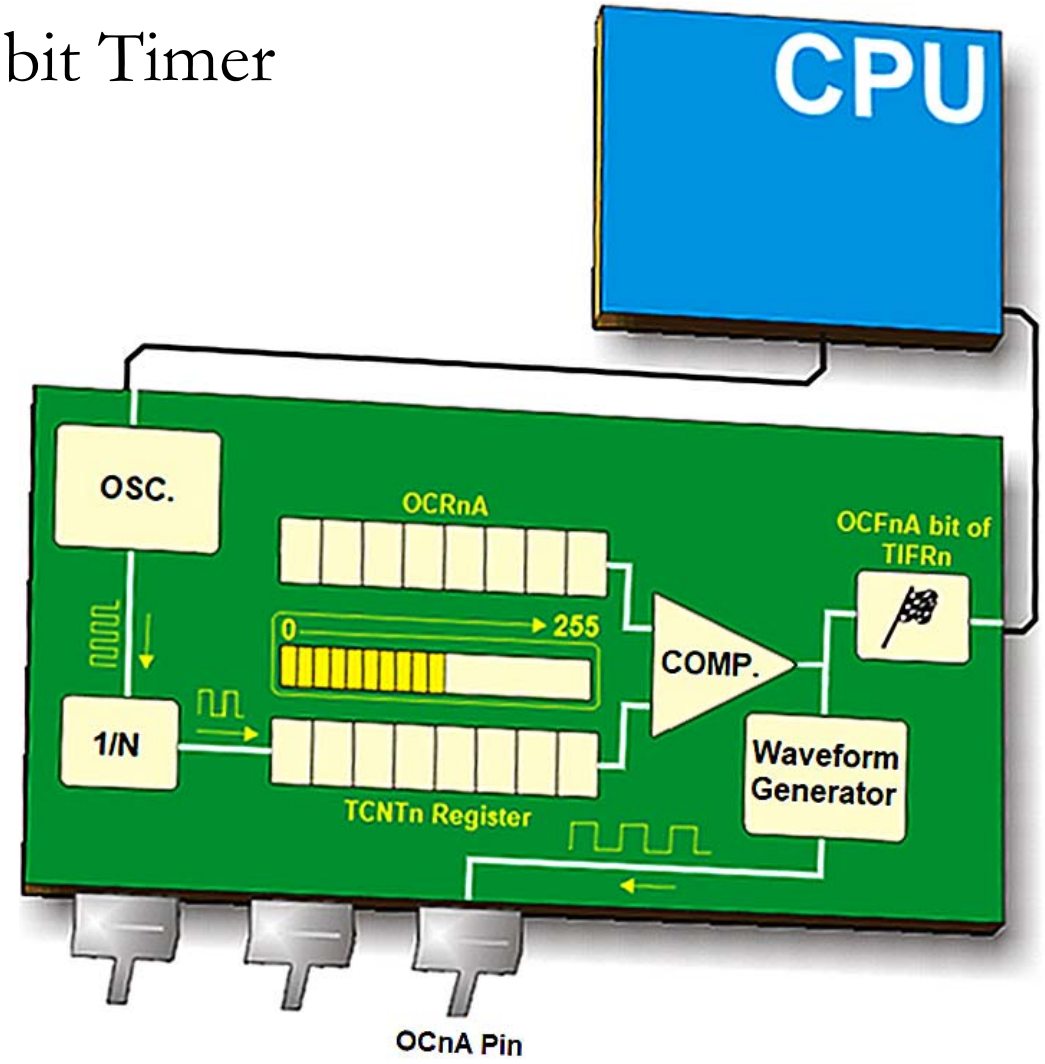




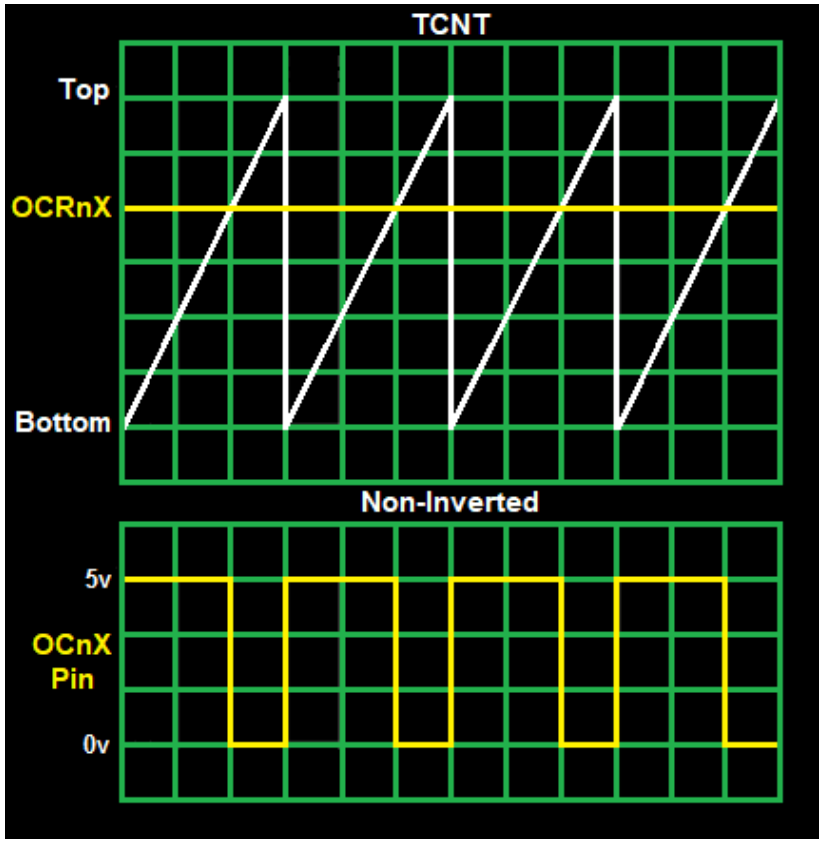


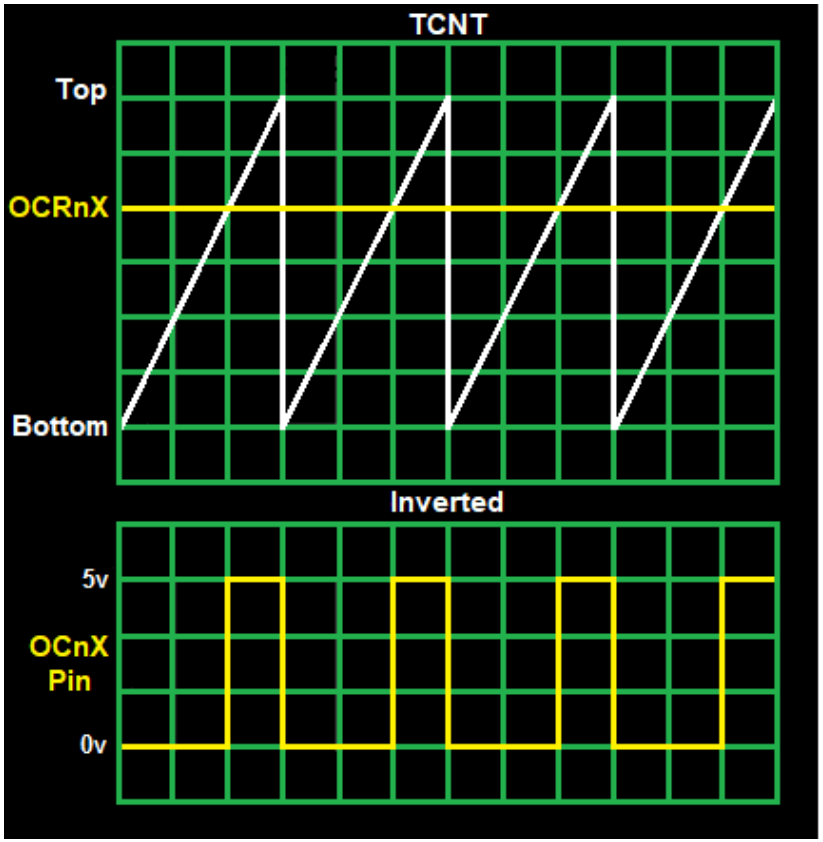
PWM

- Using 8-bit Timer



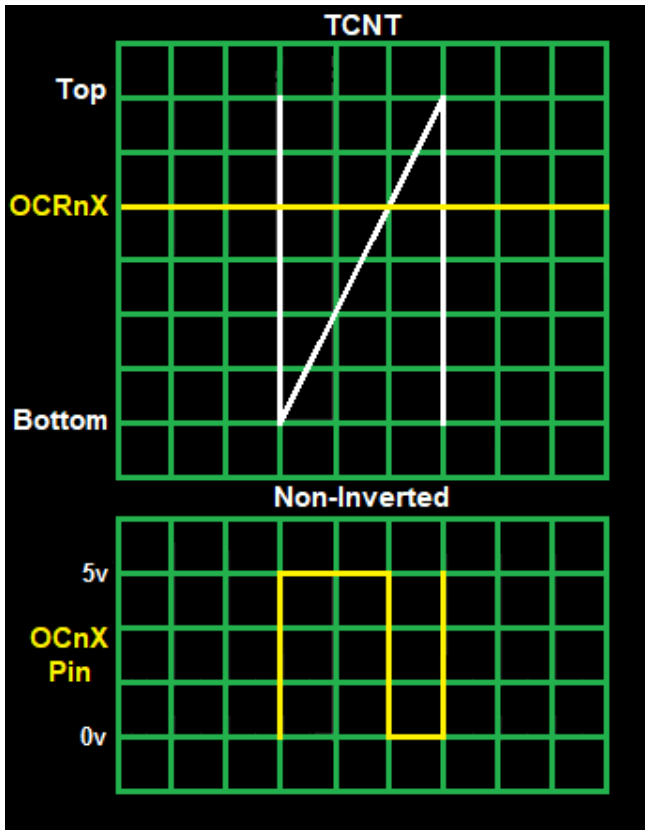
Waveform Generation





PWM Frequency

- In Non-Inverted mode, while Timer counts, if a compare match occurs, Waveform Generator clears OCnX pin and sets it when Timer comes to Bottom
- Waveform Generator generates a one PWM cycle



$$F_Timer = \frac{F_CPU}{prescale}$$

$$F_PWM = \frac{F_Timer}{(top + 1)} = \frac{F_CPU}{(top + 1) \times prescale}$$

Timer / Counter Registers

- TCCRnB (Timer/Counter Control Register n B)



  - CSn2 (Clock Select), CSn1 and CSn0 bits select prescale

CSn2	CSn1	CSn0	Clock Source
0	0	0	Disconnect clock source
0	0	1	F_CPU
0	1	0	F_CPU/8
0	1	1	F_CPU/64
1	0	0	F_CPU/256
1	0	1	F_CPU/1024

Timer / Counter Registers

- TCCRnA (Timer/Counter Control Register n A)



  - WGMn1 (Waveform Generation Mode) and WGMn0 bits with WGMn2 bit of TCCRnB specify waveform generation mode

WGMn2	WGMn1	WGMn0	Mode
0	0	0	Normal
0	0	1	Phase Correct PWM
0	1	0	Clear Timer on Compare (CTC)
0	1	1	Fast PWM

Timer / Counter Registers

- TCCRnA (Timer/Counter Control Register n A)



  - COMnA1 (Compare Output Mode), COMnA0, COMnB1 and COMnB0 bits specify compare output mode

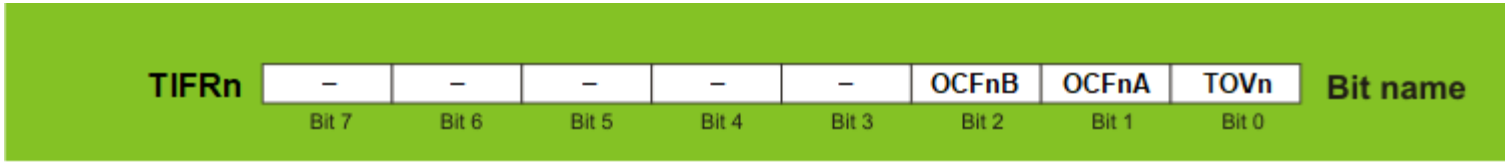
COMnX1	COMnX0	Mode	Description
1	0	Non-Inverted	Clears OCnX on compare match & sets it at Bottom
1	1	Inverted	Sets OCnX on compare match & clears it at Bottom

Timer / Counter Registers

- OCRnA (Output Compare Register n A) and OCRnB
  - hold values to be compared against TCNTn register
  - If 8-bit, can hold a value between 0 and $2^8 - 1$
  - If 16-bit, can hold a value between 0 and $2^{16} - 1$

Timer / Counter Registers

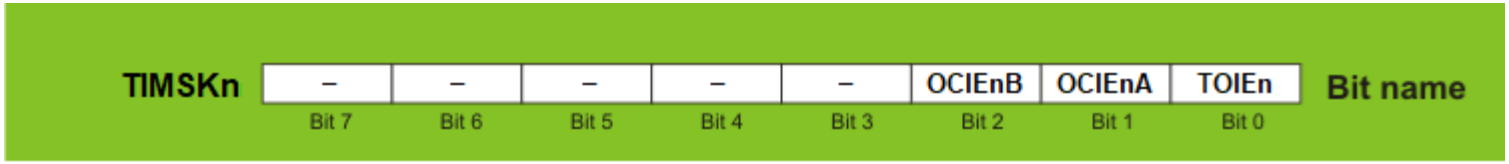
- TIFRn (Timer Interrupt Flag Register n)



  - When a compare match occurs between OCRnX and TCNTn registers, OCFnX (Output Compare match Flag n X) flag bit is set
  - Set OCFnX flag bit to clear it

Timer / Counter Registers

- TIMSKn (Timer Interrupt MaSK Register n)

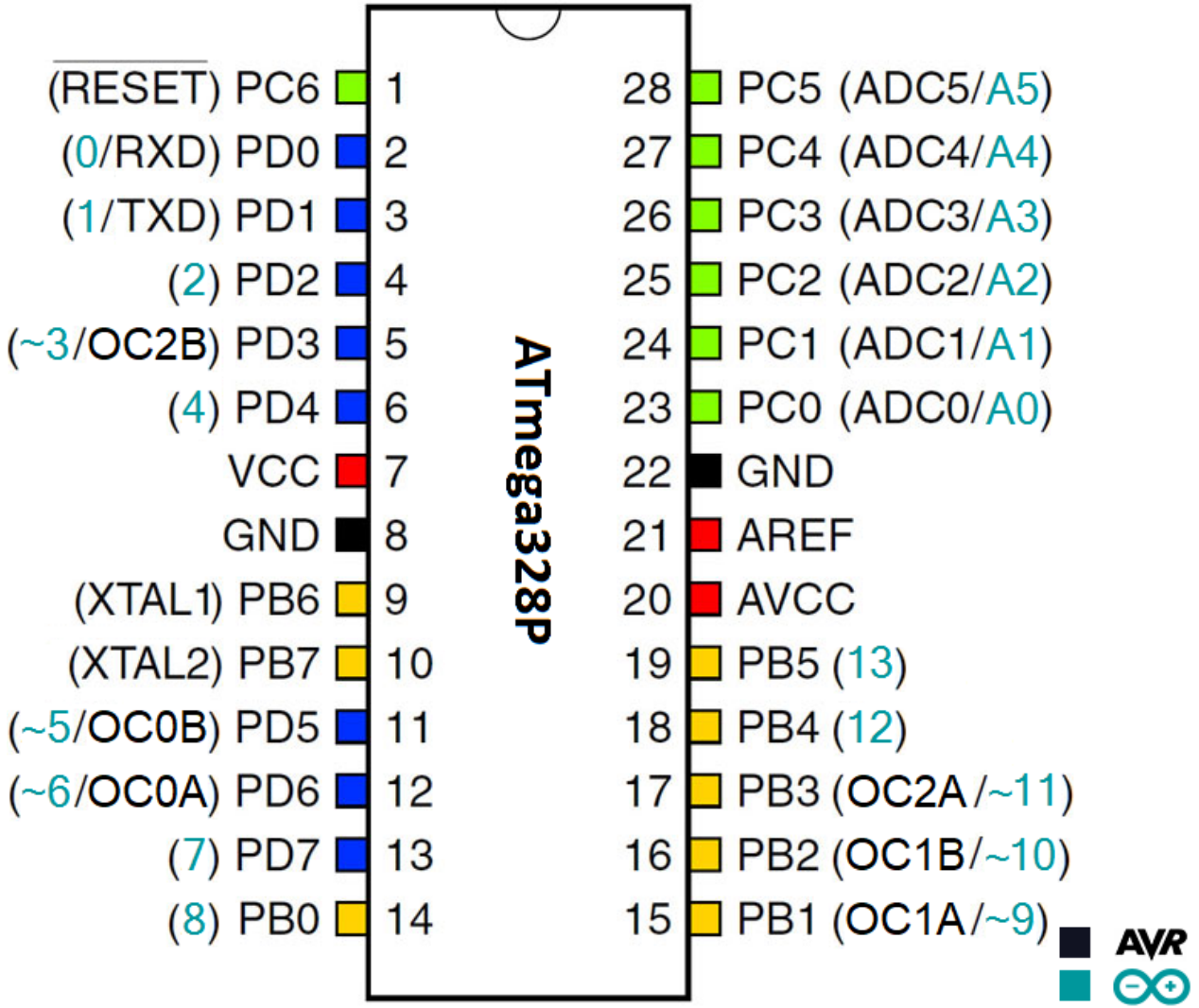


  - OCIEnA (Output Compare match Interrupt Enable n A) and OCIEnB bits enable (or disables) Timer/Counter compare match A and B interrupts, respectively

System 15

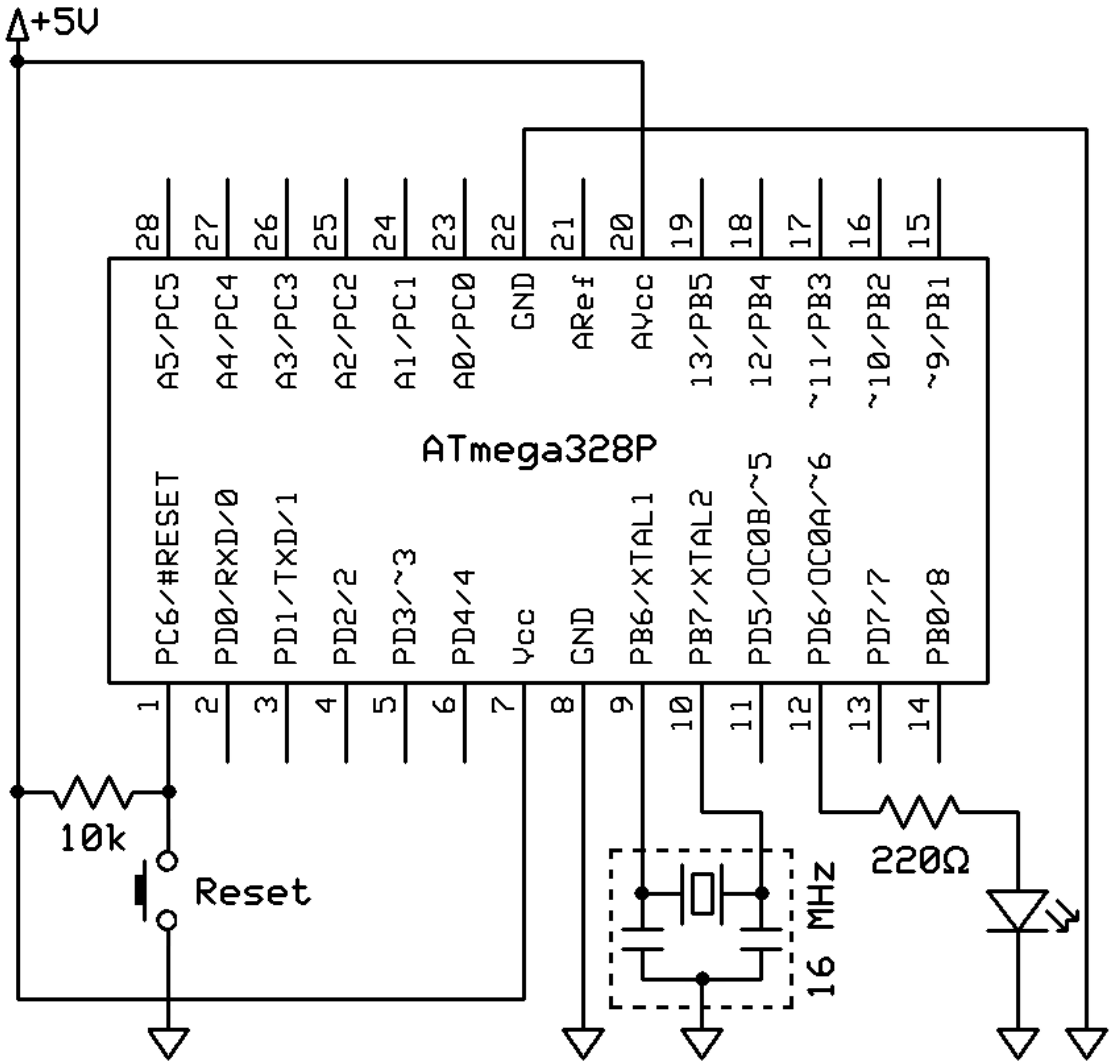
fader - Gradually fades the light of an LED again and again

ATmega328P PWM Pinout



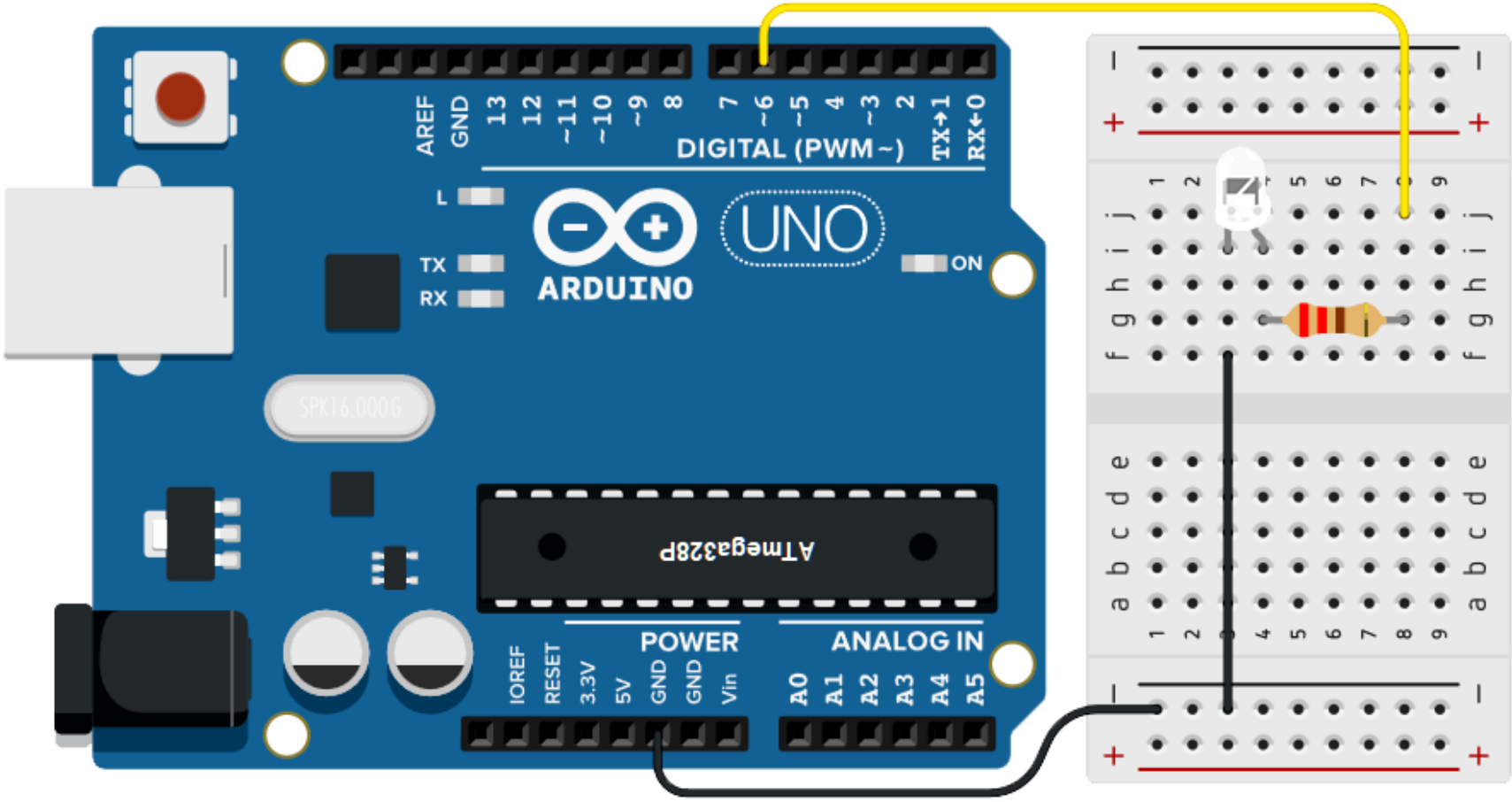
Hardware

- Schematic Diagram



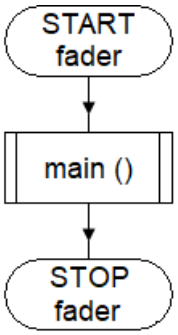
Hardware

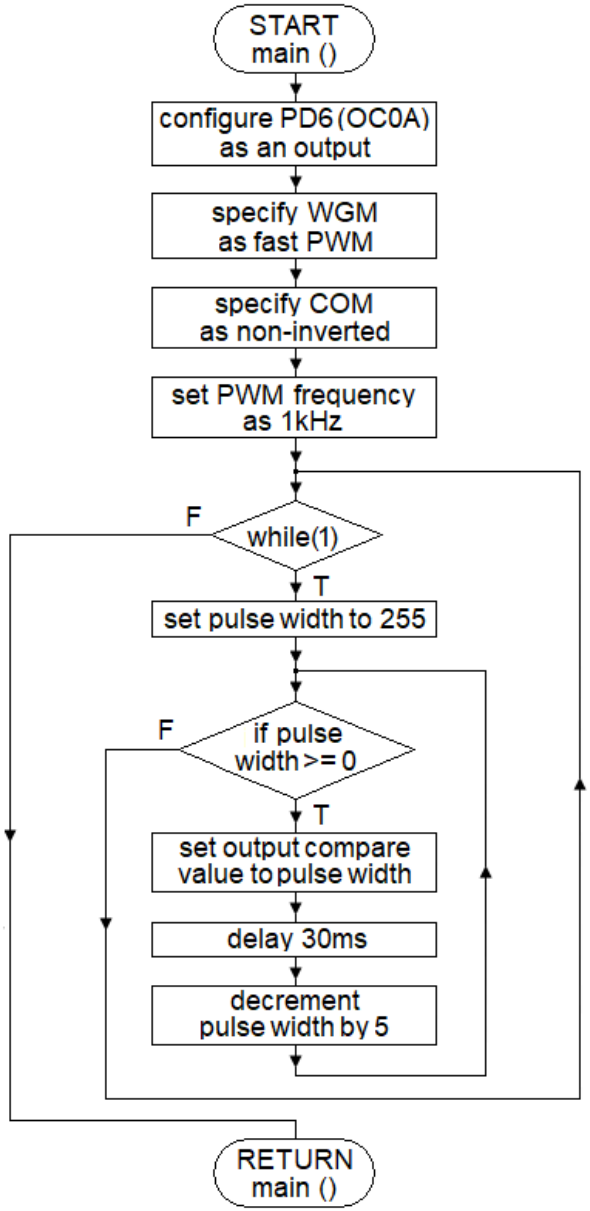
- Wiring Diagram



Firmware

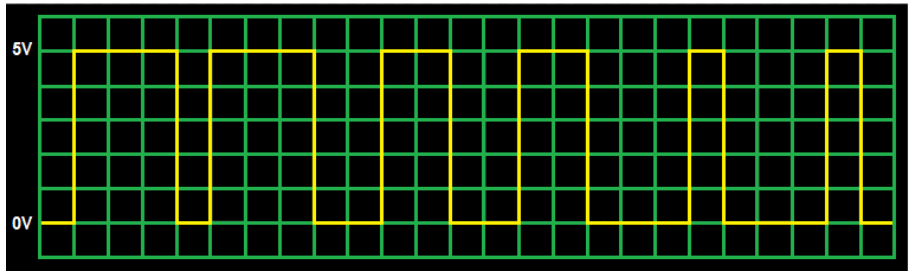
- Algorithm





C Program

```
1   // faderC.c
2
3   #include <avr/io.h>                                          // IO definitions
4   #define F_CPU 16000000UL                                     // define CPU clock speed for delay.h
5   #include <util/delay.h>                                      // definition of _delay_ms()
6
7   int main(void)
8   {
9       DDRD = DDRD | 0b01000000;                                // configure PD6 (OC0A) pin as an output
10      TCCR0B = TCCR0B & 0b11110111;                            //
11      TCCR0A = TCCR0A | 0b00000011;                            // specify WGM as fast PWM
12      TCCR0A = (TCCR0A & 0b10111111) | 0b10000000;             // specify COM as non-inverted
13      TCCR0B = (TCCR0B & 0b11111011) | 0b00000011;             // set PWM frequency as 1kHz
14      while (1)
15      {
16          for(int pulseWidth = 255; pulseWidth >= 0; pulseWidth -= 5)
17          {
18              OCR0A = pulseWidth;                              // set output compare value
19              _delay_ms(30);                                   // delay 30ms
20          }
21      }
22      return 0;
23  }
```



Using Hexadecimal Constants

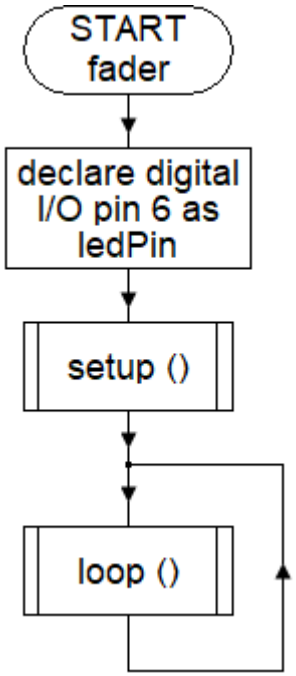
```
1   // faderC.c
2
3   #include <avr/io.h>                                    // IO definitions
4   #define F_CPU 16000000UL                               // define CPU clock speed for delay.h
5   #include <util/delay.h>                                // definition of _delay_ms()
6
7   int main(void)
8   {
9       DDRD = DDRD | 0x40;                                // configure PD6 (OC0A) pin as an output
10      TCCR0B = TCCR0B & 0xF7;                            //
11      TCCR0A = TCCR0A | 0x03;                            // specify WGM as fast PWM
12      TCCR0A = (TCCR0A & 0xBF) | 0x80;                   // specify COM as non-inverted
13      TCCR0B = (TCCR0B & 0xFB) | 0x03;                   // set PWM frequency as 1kHz
14      while (1)
15      {
16          for(int pulseWidth = 255; pulseWidth >= 0; pulseWidth -= 5)
17          {
18              OCR0A = pulseWidth;                        // set output compare value
19              _delay_ms(30);                             // delay 30ms
20          }
21      }
22      return 0;
23  }
```

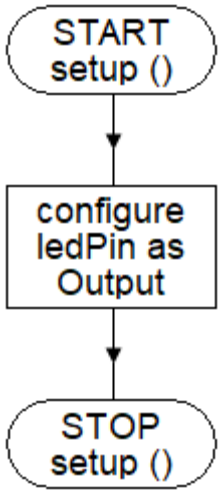
Using Labeled Constants

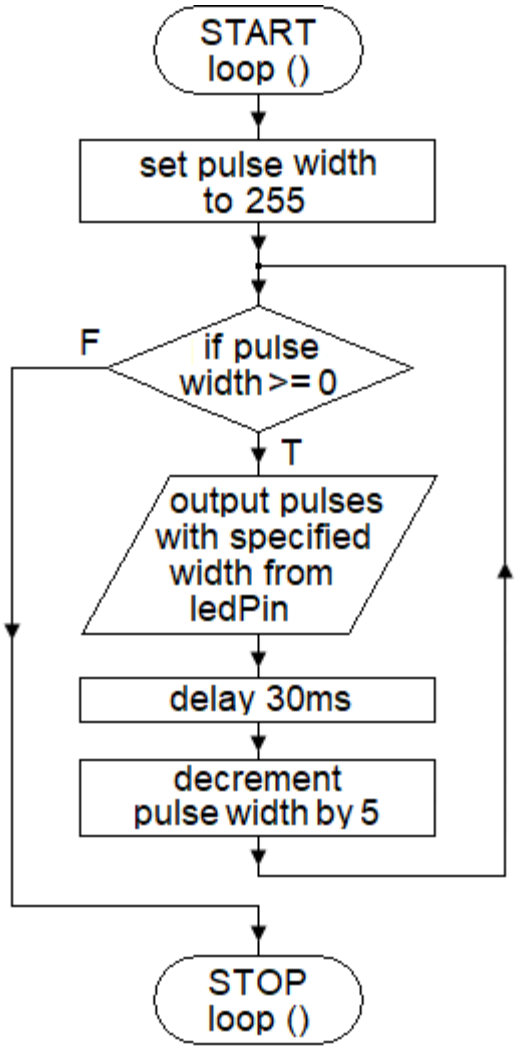
```
1   // faderC.c
2
3   #include <avr/io.h>                                          // IO definitions
4   #define F_CPU 16000000UL                                     // define CPU clock speed for delay.h
5   #include <util/delay.h>                                      // definition of _delay_ms()
6
7   int main(void)
8   {
9       DDRD = DDRD | (1<<DDD6);                                 // configure PD6 (OC0A) pin as an output
10      TCCR0B = TCCR0B & ~(1<<WGM02);                           //
11      TCCR0A = TCCR0A | (1<<WGM01) | (1<<WGM00);               // specify WGM as fast PWM
12      TCCR0A = (TCCR0A & ~(1<<COM0A0)) | (1<<COM0A1);          // specify COM as non-inverted
13      TCCR0B = (TCCR0B & ~(1<<CS02)) | (1<<CS01) | (1<<CS00);  // set PWM frequency as 1kHz
14      while (1)
15      {
16          for(int pulseWidth = 255; pulseWidth >= 0; pulseWidth -= 5)
17          {
18              OCR0A = pulseWidth;                              // set output compare value
19              _delay_ms(30);                                   // delay 30ms
20          }
21      }
22      return 0;
23  }
```

Firmware

- Algorithm for Arduino







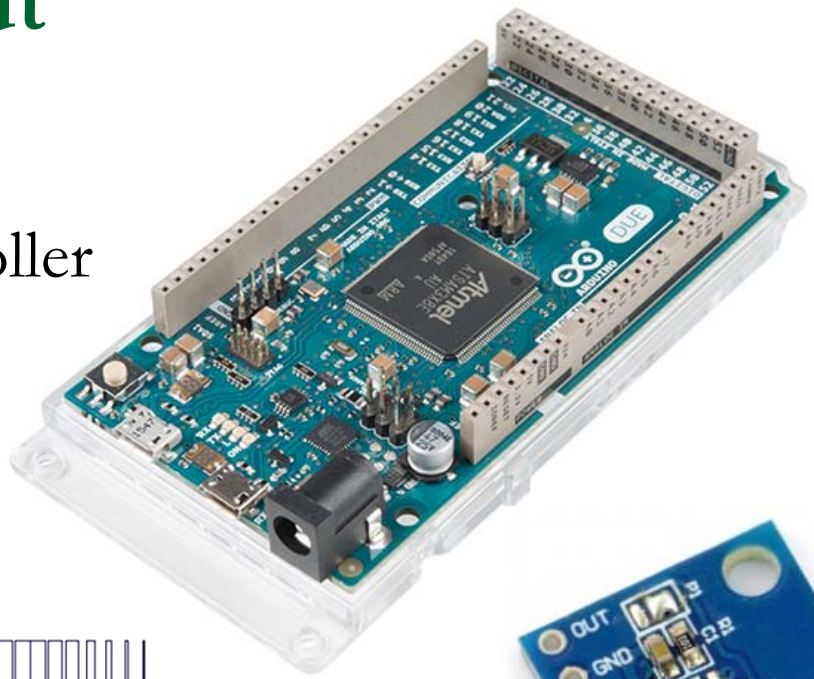
Firmware

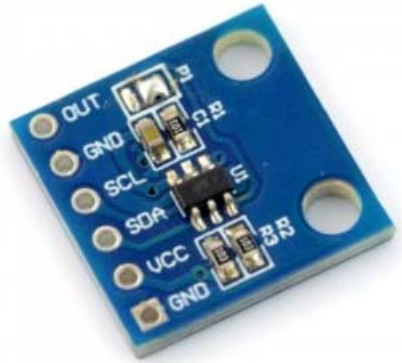
- Source Code (using Arduino Wrapper Functions)

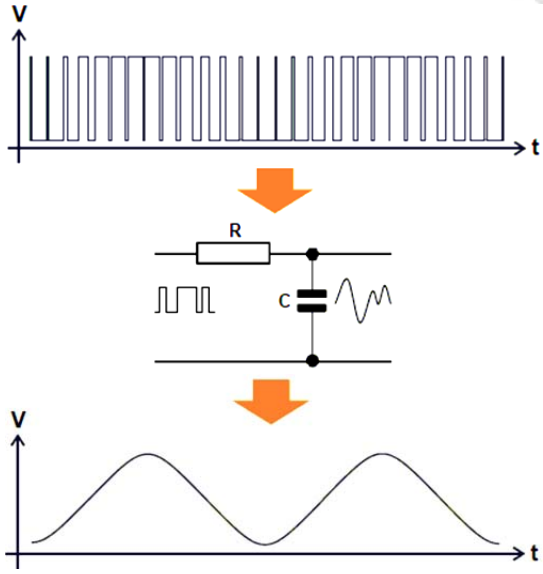
```
1 //fader.ino
2
3 int ledPin = 6;                          // declare digital I/O pin 6 as ledPin
4
5 void setup()
6 {
7   pinMode(ledPin, OUTPUT);              // configure LED connected pin as output
8 }
9
10 void loop()
11 {
12   for(int pulseWidth = 255; pulseWidth >=0; pulseWidth -= 5)
13   {
14     analogWrite(ledPin, pulseWidth); // output pulses with specified width
15     delay(30);                        // wait for 30ms
16   }
17 }
```

True Analog Output

- Three options
  - Switch to a microcontroller with built-in DAC
  - Add a DAC breakout
  - Use a RC filter







Questions